

1 Preliminary Knowledge

1.1 Quadrature Operator

We define the quadrature operator $\hat{X}_\theta$ as

$$\hat{X}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta} \quad (1)$$

The conventional field quadratures used in quantum optics are $\hat{X} = \hat{X}_0$ and $\hat{P} = \hat{X}_{\pi/2}$

$$\hat{X} = \hat{a} + \hat{a}^\dagger, \quad \hat{P} = (\hat{a} - \hat{a}^\dagger)/i \quad (2)$$

and satisfy the commutation relation

$$[\hat{X}, \hat{P}] = 2i, \quad (3)$$

building upon the fact that $[\hat{a}, \hat{a}^\dagger] = 1$. These field quadratures define the Wigner function plane.

1.2 Fock States

In the Fock basis representation, Fock (or number) states are denoted as $|n\rangle$, where n is the number of photons (or excitations) in the vacuum $|0\rangle$. They are eigenstates of the number operator

$$\hat{n}|n\rangle = n|n\rangle, \quad (\hat{n} = \hat{a}^\dagger \hat{a}). \quad (4)$$

As eigenstates of the number operator, they have a well-defined number of photons $(\Delta\hat{n})^2 = \langle\hat{n}^2\rangle - \langle\hat{n}\rangle^2 = 0$, but an undefined optical phase ϕ . Their wigner function is a gaussian centered at the origin.

In the position basis representation, a fock state is written as

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\pi^{1/4}\sqrt{2^n n!}} H_n(x) e^{-x^2/2}, \quad (5)$$

where $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ is the Hermite polynomial of order n .

1.3 Coherent States

Coherent states are defined as the eigenstates of the annihilation operator

$$\hat{a}|a\rangle = a|a\rangle, \quad a \in \mathbb{C}. \quad (6)$$

The complex amplitude of the state can be written as $\langle\hat{a}\rangle = a = |a|e^{i\phi}$, where ϕ is the optical phase of the electromagnetic field. In the Fock basis, a coherent state is decomposed as

$$|a\rangle = e^{-\frac{|a|^2}{2}} \sum_{n=0}^{\infty} \frac{a^n}{\sqrt{n!}} |n\rangle. \quad (7)$$

The average number of photons in a coherent state is

$$\langle a|\hat{n}|a\rangle = |a|^2. \quad (8)$$

Such state can be generated through displacement of the vacuum

$$|a\rangle = \hat{D}(a)|0\rangle, \quad \hat{D}(a) = \exp[a\hat{a}^\dagger - a^*\hat{a}]. \quad (9)$$

In our convention, the Wigner function of a coherent state is displaced from the origin by an amount $2|a|$ and at an angle ϕ from the x-axis.

1.4 Squeezed States

Squeezed states are generated through squeezing of the vacuum

$$|z\rangle = \hat{S}(z) |0\rangle, \quad \hat{S}(z) = \exp \left[\frac{1}{2} (z \hat{a}^2 - z^* \hat{a}^{\dagger 2}) \right]. \quad (10)$$

The complex number $z = |r|e^{i\theta}$ determines the squeezing parameter r and the squeezed quadrature $\hat{X}_\theta$. Squeezing the quadrature $\hat{X}_\theta$ results in the anti-squeezing of quadrature $\hat{X}_{\theta+\pi/2}$

$$(\Delta \hat{X}_\theta)^2 \propto e^{-2r}, \quad (\Delta \hat{X}_{\theta+\pi/2})^2 \propto e^{2r}. \quad (11)$$

The average photon number of a squeezed state is

$$\langle z | \hat{n} | z \rangle = \sinh^2(r). \quad (12)$$

The Wigner function is elliptical due to the squeezing, and at an angle $\theta/2$ with respect to the x-axis.

1.5 Displaced Squeezed States (DSS)

Such states are generated through squeezing and displacement of the vacuum

$$|a, z\rangle = \hat{D}(a) \hat{S}(z) |0\rangle. \quad (13)$$

Previous points for displaced and squeezed states hold independently. The average number of photons in a DSS is

$$\langle a, z | \hat{n} | a, z \rangle = |a|^2 + \sinh^2(r). \quad (14)$$

The Wigner function is again elliptical, displaced from the origin by an amount $2|a|$, is at an angle ϕ with respect to the x-axis, and rotated by $\theta/2$.

2 Discrimination between pure Quantum States in a communication channel

The minimum theoretical error probability of mis-discriminating between two non-orthogonal pure states $|\psi_{1,2}\rangle$ (with equal prior probability of being selected) is given by the Helstrom bound

$$P^{(\min)}(|\psi_1\rangle, |\psi_2\rangle) = \frac{1}{2} \left(1 - \sqrt{1 - |\langle \psi_1 | \psi_2 \rangle|^2} \right) \quad (15)$$

Experimentally, one cannot reach the absolute minimum probability of making an error when trying to discriminate between two states. In this work, Homodyne detection is considered as the experimental means of a receiver to discriminate between two states, which clearly leads to a higher error probability. The strategy used by the receiver is the following:

$$\begin{cases} x \geq 0 \Rightarrow |0\rangle \equiv |\psi_1\rangle, \\ x < 0 \Rightarrow |1\rangle \equiv |\psi_2\rangle \end{cases} \quad (16)$$

where x is drawn from the Homodyne detection probability distribution

$$x \sim P^{(\text{HD})}(x; |\psi_{1,2}\rangle) = |\langle x | \psi_{1,2} \rangle|^2, \quad (17)$$

with $\hat{X}|x\rangle = x|x\rangle$. The probability of making an error is given by

$$P_{\text{err}} = \frac{1}{2} [p(x \geq 0|x=1) + p(x < 0|x=0)] = p(x \geq 0|x=1); \quad p(x \geq 0|x=1) = p(x < 0|x=0) \quad (18)$$

The conditional probability is given by

$$p(x \geq 0|x=1) = \int_0^\infty P^{(\text{HD})}(x; |\psi_{1,2}\rangle) dx. \quad (19)$$

To approximate this probability, we draw thousands of samples from $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$, and define the error probability as

$$p(x \geq 0|x=1) = P_{\text{err}} = \frac{\text{\# of samples where } x \geq 0}{\text{\# of all samples}} \quad (20)$$

$$p(x < 0|x=0) = P_{\text{err}} = \frac{\text{\# of samples where } x < 0}{\text{\# of all samples}} \quad (21)$$

The goal of the work is to determine whether the application of certain combinations of quantum resources on the states (e.g. displacement, squeezing) can make $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$ approach $P^{(\min)}(|\psi_1\rangle, |\psi_2\rangle)$ as much as possible with respect to other combinations.

2.1 Discrimination between Coherent States

In the case of coherent state communication our states are defined as

$$|\psi_\pm\rangle = \hat{D}(\pm a)|0\rangle, \quad (22)$$

where $\pm a \in \mathbb{R}$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{\text{CS}}^{(\min)}(N) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp(-4N)}, \quad (23)$$

where N is the average number of photons in the coherent states. The homodyne detection probability distribution becomes

$$P_{\text{CS}}^{(\text{HD})}(x|\pm) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(x \mp 2\sqrt{N})^2}{2} \right], \quad (24)$$

with the error probability being

$$P_{\text{err}}^{CS}(N) = \int_0^\infty P_{\text{CS}}^{(\text{HD})}(x|-) dx = \boxed{\frac{1}{2} \text{erfc}(\sqrt{2N})}. \quad (25)$$

2.2 Discrimination between DSS

In the case of communication with DSS, our states are defined as

$$|\psi_\pm\rangle = \hat{D}(\pm a) \hat{S}(r) |0\rangle, \quad (26)$$

where $z = r \in \mathbb{R}$ and $\pm a$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{\text{DSS}}^{(\text{min})}(N, r) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp(-4Ne^{-2r})}, \quad (27)$$

where N is the average number of photons in the DSS

$$N = |a|^2 + \sinh^2(r). \quad (28)$$

We may define the squeezing fraction $\beta \equiv \sinh^2(r)/N$, and express the relevant equations with respect to that. The minimum error probability (Helstrom Bound) is given by

$$P_{\text{DSS}}^{(\text{min})}(N, \beta) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp \left[-4N(1 - \beta) \times \left(1 + 2N\beta + 2\sqrt{N\beta(1 + N\beta)} \right) \right]}, \quad (29)$$

The homodyne detection probability distribution becomes

$$P_{\text{DSS}}^{(\text{HD})}(x|\pm) = \frac{\Sigma}{\sqrt{2\pi}} \exp \left[-\frac{1}{2} \Sigma^2 (x \mp 2a)^2 \right], \quad (30)$$

with

$$\alpha = \sqrt{N(1 - \beta)}, \quad \Sigma = \left(\sqrt{N\beta} + \sqrt{1 + N\beta} \right). \quad (31)$$

The error probability is given by

$$P_{\text{err}}^{\text{DSS}}(N, \sigma, \mu) = \int_0^\infty P_{\text{DSS}}^{(\text{HD})}(x|-) dx = \boxed{\frac{1}{2} \text{erfc}(a\Sigma)}. \quad (32)$$

2.3 Which is easier to discriminate, Coherent states or DSS?

For a certain fraction of squeezing, DSS can outperform Coherent states. More specifically, for a fixed number of photons $\tilde{N}$, DSS have a lower error probability if β is below a threshold β_{th}

$$P_{\text{err}}^{\text{DSS}}(\beta|\tilde{N}) = P_{\text{err}}^{\text{CS}}(\tilde{N}) \quad \text{if } \beta = \beta_{\text{th}}(\tilde{N}), \quad (33)$$

where

$$\beta_{\text{th}}(\tilde{N}) = \frac{4\tilde{N}}{4\tilde{N} + 1}. \quad (34)$$

Additionally, for a fixed number of photons $\tilde{N}$, there exists an optimal fraction of squeezing $\beta_{\text{opt}} \in [0, \beta_{\text{th}}]$ for which the error probability is minimized

$$\left. \frac{\partial P_{\text{err}}^{\text{DSS}}}{\partial \beta} \right|_{\beta_{\text{opt}}, \tilde{N}} = 0 \Rightarrow \beta_{\text{opt}} = \frac{\tilde{N}}{2\tilde{N} + 1}. \quad (35)$$

Summarizing, for a fixed number of photons $\tilde{N}$, DSS will always outperform CS if the fraction of squeezing is lower than $\beta_{\text{th}}(\tilde{N})$. Moreover, the error probability will be minimized if the fraction of squeezing is equal to $\beta_{\text{opt}}(\tilde{N})$.

3 Discrimination between Quantum States under phase diffusion in a communication channel

We can describe the effect of phase diffusion on the state through the following differential equation for the density matrix

$$\frac{1}{\Gamma} \frac{d\rho(t)}{dt} = 2\hat{n}\rho\hat{n}^\dagger - \hat{n}^\dagger\hat{n}\rho - \rho\hat{n}^\dagger\hat{n}; \quad (36)$$

we may write the density operator in the Fock basis by using the resolution of identity

$$\rho = \mathbb{I}\rho\mathbb{I} = \left(\sum_n |n\rangle \langle n| \right) \rho \left(\sum_m |m\rangle \langle m| \right), \quad (37)$$

$$= \sum_{n,m} \langle n|\rho|m\rangle |n\rangle \langle m|, \quad (38)$$

$$= \sum_{n,m} \rho_{n,m} |n\rangle \langle m|. \quad (39)$$

Then, eq. (36) becomes

$$\frac{1}{\Gamma} \frac{d\rho(t)}{dt} = 2 \sum_{n,m} nm \rho_{n,m} |n\rangle \langle m| - \sum_{n,m} n^2 \rho_{n,m} |n\rangle \langle m| - \sum_{n,m} m^2 \rho_{n,m} |n\rangle \langle m|, \quad (40)$$

$$\Rightarrow \sum_{n,m} \left[\frac{1}{\Gamma} \dot{\rho}_{n,m} - 2nm\rho_{n,m} + (n^2 + m^2)\rho_{n,m} \right] |n\rangle \langle m| = 0, \quad (41)$$

$$\Rightarrow \dot{\rho}_{n,m} + \Gamma(n-m)^2 \rho_{n,m} = 0. \quad (42)$$

Finally,

$$\rho_{n,m}(t) = e^{-(n-m)^2 \Gamma t} \rho_{n,m}(0); \quad (43)$$

the full density matrix is

$$\rho(t) = \sum_{n,m} \rho_{n,m}(t) |n\rangle \langle m| = \sum_{n,m} e^{-(n-m)^2 \Gamma t} \rho_{n,m}(0) |n\rangle \langle m|. \quad (44)$$

By the use of Fourier transform theory, we can rewrite the above equation as

$$\rho(t) = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \hat{U}_\phi \rho(0) \hat{U}_\phi^\dagger d\phi, \quad (45)$$

where $\hat{U}_\phi = e^{-i\hat{n}\phi}$ and $\sigma \propto \sqrt{\Gamma}$ is the noise parameter. This operator has the following effect on Coherent states and DSS

$$\hat{U}_\phi |\pm a\rangle = e^{-i\hat{n}\phi} |\pm a\rangle = |\pm e^{-i\phi} a\rangle, \quad (46)$$

$$\hat{U}_\phi |\pm a, r\rangle = e^{-i\hat{n}\phi} |\pm a, r\rangle = |\pm e^{-i\phi} a, e^{-i2\phi} r\rangle. \quad (47)$$

Evidently, this operator produces an optical phase rotation. So, for both Coherent and DSS, the density matrix of eq. (88) is

$$\rho^{CS,DSS}(t) = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \times \left(\rho_{\text{phase-shift}}^{CS,DSS} \right) d\phi \quad (48)$$

$$= \int_{-\infty}^{+\infty} P(\phi) \times \left(\rho_{\text{phase-shift}}^{CS,DSS} \right) d\phi \quad (49)$$

Thus, the state is a statistical mixture of phase-rotated Coherent states or DSS, where the phase is drawn from a Gaussian distribution with variance σ^2 . This is not the same as having a superposition of phase-rotated states; the state has undergone only one specific phase rotation - no one knows which exactly, this is what makes the state mixed. Thus, to reproduce the state in simulation, one needs to create a pure Coherent state / DSS, and then apply a phase rotation, where the phase is drawn from a Gaussian distribution with mean zero and variance σ^2

$$\rightarrow \text{create desired state } |\psi\rangle = \hat{D}(\pm a)\hat{S}(r)|0\rangle, \quad (50)$$

$$\rightarrow \text{pick } \phi \sim \mathcal{N}(0, \sigma^2), \quad (51)$$

$$\rightarrow \text{apply phase shift operator } \hat{U}_\phi |\psi\rangle. \quad (52)$$

Then, Homodyne measurement is carried out as originally.

3.1 Discrimination between phase diffused Coherent states

In the case of coherent state communication our states are defined as

$$\rho_\pm = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) |\pm e^{-i\phi}a\rangle \langle \pm e^{-i\phi}a| d\phi, \quad (53)$$

where $\pm a \in \mathbb{R}$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{CS}^{(\min)}(\rho_+, \rho_-) = \frac{1}{2} \left[1 - \frac{1}{2} \text{Tr} \left[\sqrt{(\rho_+ - \rho_-)^\dagger (\rho_+ - \rho_-)} \right] \right]. \quad (54)$$

The Homodyne detection probability distribution is

$$P_{CS}^{(\text{HD})}(x|\pm) = \int_{-\infty}^{+\infty} P(\phi) |\langle x|\pm e^{i\phi}a\rangle|^2 d\phi, \quad (55)$$

$$= \int_{-\infty}^{+\infty} \frac{P(\phi)}{\sqrt{2\pi}} \exp \left[-\frac{(x \mp 2a \cos \phi)^2}{2} \right] d\phi, \quad (56)$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(x \mp 2\sqrt{N} \cos \phi)^2}{2} \right] d\phi, \quad (57)$$

with the error probability being

$$P_{\text{err}}^{CS}(N, \sigma) = \int_0^\infty P_{CS}^{(\text{HD})}(x|-) dx = \boxed{\frac{1}{2} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \text{erfc} \left(\sqrt{2N} \cos \phi \right) d\phi}. \quad (58)$$

3.2 Discrimination between phase diffused DSS

In the case of communication with DSS our states are defined as

$$\rho_\pm = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) |\pm e^{-i\phi}a, e^{-i2\phi}r\rangle \langle \pm e^{-i\phi}a, e^{-i2\phi}r| d\phi, \quad (59)$$

where $z = r \in \mathbb{R}$ and $\pm a$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{DSS}^{(\min)}(\rho_+, \rho_-) = \frac{1}{2} \left[1 - \frac{1}{2} \text{Tr} \left[\sqrt{(\rho_+ - \rho_-)^\dagger (\rho_+ - \rho_-)} \right] \right]. \quad (60)$$

The Homodyne detection probability distribution is

$$P_{DSS}^{(\text{HD})}(x|\pm) = \int_{-\infty}^{+\infty} P(\phi) |\langle x | \pm e^{i\phi} a, e^{-i2\phi} r \rangle|^2 d\phi, \quad (61)$$

$$= \int_{-\infty}^{+\infty} \frac{P(\phi)}{\sqrt{2\pi V(\phi, \beta, N)}} \exp \left[-\frac{(x \mp 2\sqrt{N(1-\beta)} \cos \phi)^2}{2V(\phi, \beta, N)} \right] d\phi, \quad (62)$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi V(\phi, \beta, N)}} \exp \left[-\frac{(x \mp 2\sqrt{N(1-\beta)} \cos \phi)^2}{2V(\phi, \beta, N)} \right] d\phi, \quad (63)$$

where

$$V(\phi, \beta, N) = \exp \left[-2 \sinh^{-1} \left(\sqrt{N\beta} \right) \right] \cos^2 \phi + \exp \left[2 \sinh^{-1} \left(\sqrt{N\beta} \right) \right] \sin^2 \phi. \quad (64)$$

The error probability is

$$P_{\text{err}}^{\text{DSS}}(N, \sigma, \beta) = \int_0^\infty P_{DSS}^{(\text{HD})}(x|-) dx = \boxed{\frac{1}{2} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \text{erfc} \left(\frac{2\sqrt{N(1-\beta)} \cos \phi}{\sqrt{2V(\phi, \beta, N)}} \right) d\phi}. \quad (65)$$

3.3 Which is easier to discriminate, Coherent states or DSS?

The amount of squeezing fraction under which squeezing aids discrimination satisfies the following condition (solved numerically)

$$P_{\text{err}}^{\text{DSS}}(N, \sigma | \beta_{\text{th}}) - P_{\text{err}}^{\text{CS}}(N, \sigma) = 0. \quad (66)$$

The optimal squeezing fraction β_{opt} is given by

$$\beta_{\text{opt}}(\tilde{N}) = \min_{\beta} P_{DSS}^{(\text{HD})}(\beta | \tilde{N}), \quad (67)$$

which is also solved numerically.

4 Discrimination between Quantum States under phase diffusion and impurity in a communication channel

Previously, our initial pure state was a displaced squeezed vacuum

$$\rho(0) = \hat{D}(\pm a) \hat{S}(r) |0\rangle \langle 0| \hat{S}^\dagger(r) \hat{D}^\dagger(\pm a), \quad (68)$$

where $\text{Tr}[\rho(0)^2] = 1$. Now, we introduce an impurity by assuming initially the existence of a displaced squeezed thermal state $v(\mu)$ of purity $\mu = \text{Tr}[\rho(0)^2]$, where

$$\rho(0) = \hat{D}(\pm a) \hat{S}(r) v(\mu) \hat{S}^\dagger(r) \hat{D}^\dagger(\pm a). \quad (69)$$

The thermal state is given by

$$v(\mu) = \frac{2\mu}{1+\mu} \sum_{n=0}^{\infty} \left(\frac{1-\mu}{1+\mu} \right)^n |n\rangle \langle n|, \quad (70)$$

with mean photon number

$$N_{\text{thermal}} = \text{Tr}[v(\mu) \hat{n}] = \frac{2\mu}{1+\mu} \sum_{n=0}^{\infty} \left(\frac{1-\mu}{1+\mu} \right)^n \text{Tr}[|n\rangle \langle n| \hat{n}] \quad (71)$$

$$= \frac{2\mu}{1+\mu} \sum_{n=0}^{\infty} \left(\frac{1-\mu}{1+\mu} \right)^n \underbrace{\sum_m \langle m|n\rangle \langle n|\hat{n}|m\rangle}_{m\delta_{mn}=n} \quad (72)$$

$$= \frac{2\mu}{1+\mu} \sum_{n=0}^{\infty} n \left(\frac{1-\mu}{1+\mu} \right)^n \quad (73)$$

$$= \frac{1-\mu}{2\mu}. \quad (74)$$

The entire displaced squeezed thermal state (DSTS) has mean photon number (in terms of the squeezing fraction β)

$$N = |a|^2 + N\beta + \frac{1-\mu}{2\mu} (1 + 2N\beta), \quad (75)$$

where we demand that $|a|^2 \geq 0$

$$|a|^2 = N(1-\beta) + \frac{\mu-1}{2\mu} (1 + 2N\beta) \geq 0 \quad (76)$$

$$\Rightarrow \left(\frac{1}{2} + N\beta \right) + N(1-\beta) - \frac{1}{\mu} \left(\frac{1}{2} + N\beta \right) \geq 0 \quad (77)$$

$$\Rightarrow \left(\frac{1}{2} + N\beta \right) \left(1 - \frac{1}{\mu} \right) \geq N(\beta - 1) \quad (78)$$

$$\Rightarrow 1 - \frac{1}{\mu} \geq \frac{2N(\beta - 1)}{1 + 2N\beta} \quad (79)$$

$$\Rightarrow 1 - \frac{2N(\beta - 1)}{1 + 2N\beta} = \frac{1 + 2N}{1 + 2N\beta} \geq \frac{1}{\mu} \quad (80)$$

$$\Rightarrow \mu \geq \frac{1 + 2N\beta}{1 + 2N}. \quad (81)$$

The squeezing fraction β is bounded between 0 and 1 itself; taking this into account, μ is constrained in the following way for the lower bound of β

$$\mu \geq \frac{1}{1+2N} \quad (\beta = 0); \quad (82)$$

for the upper bound of $\beta = 1$, $\mu \geq 1$. But, since μ is also bounded between 0 and 1 as a measure of impurity, when all the energy is allocated to squeezing ($\beta = 1$), there is no energy left for thermal noise (or displacement), and thus the state is pure ($\mu = 1$). Taking these considerations into account, impurity μ is finally bounded in the following way

$$\frac{1}{1+2N} \leq \mu \leq 1. \quad (83)$$

The evolution of the state under phase diffusion is now given by

$$\rho(t) = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \hat{U}_\phi \rho(0) \hat{U}_\phi^\dagger d\phi \quad (84)$$

$$(85)$$

$$= \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \hat{U}_\phi \hat{D}(\pm a) \hat{S}(r) \nu(\mu) \hat{S}^\dagger(r) \hat{D}^\dagger(\pm a) \hat{U}_\phi^\dagger d\phi \quad (86)$$

$$(87)$$

$$= \int_{-\infty}^{+\infty} P(\phi) \times \left(\rho_{\text{phase-shift}}^{DSTS} \right) d\phi. \quad (88)$$

In case of no squeezing, we have got just a displaced thermal state (DTS).

Thus, **the state is a statistical mixture of phase-rotated displaced (and possibly squeezed) thermal states, where the phase is drawn from a Gaussian distribution with variance σ^2** . To reproduce the state in simulation, one needs to create a displaced (squeezed) thermal state, and then apply a phase rotation, where the phase is drawn from a Gaussian distribution with mean zero and variance σ^2

$$\rightarrow \text{create desired state } |\psi\rangle = \hat{D}(\pm a) \hat{S}(r) \nu(\mu), \quad (89)$$

$$\rightarrow \text{pick } \phi \sim \mathcal{N}(0, \sigma^2), \quad (90)$$

$$\rightarrow \text{apply phase shift operator } \hat{U}_\phi |\psi\rangle. \quad (91)$$

Then, Homodyne measurement is carried out as originally.

4.1 Discrimination between phase diffused Displaced Thermal States (DTS)

In this case our states are defined as

$$\rho_\pm = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \hat{D}(\pm a e^{-i\phi}) \nu(\mu) \hat{D}^\dagger(\pm a e^{-i\phi}) d\phi, \quad (92)$$

where $\pm a \in \mathbb{R}$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{DTS}^{(\min)}(\rho_+, \rho_-) = \frac{1}{2} \left[1 - \frac{1}{2} \text{Tr} \left[\sqrt{(\rho_+ - \rho_-)^\dagger (\rho_+ - \rho_-)} \right] \right]. \quad (93)$$

The Homodyne detection probability distribution is

$$P_{DTS}^{(HD)}(x|\pm) = \int_{-\infty}^{+\infty} P(\phi) \langle x|\hat{D}(\pm ae^{-i\phi})\nu(\mu)\hat{D}^\dagger(\pm ae^{-i\phi})|x\rangle d\phi, \quad (94)$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi/\mu}} \exp \left[-\frac{\left(x \mp \sqrt{2\left(N + \frac{\mu-1}{2\mu}\right) \cos \phi} \right)^2}{2/\mu} \right] d\phi, \quad (95)$$

with the error probability being

$$P_{\text{err}}^{(HD)}(N, \sigma, \mu) = \int_0^\infty P_{DTS}^{(HD)}(x|-) dx \quad (96)$$

$$= \left[\frac{1}{2} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \text{erfc} \left(\sqrt{\mu \left(N + \frac{\mu-1}{2\mu} \right) \cos \phi} \right) d\phi \right]. \quad (97)$$

4.2 Discrimination between phase diffused Displaced Squeezed Thermal States (DSTS)

In this case our states are defined as

$$\rho_\pm = \int_{-\infty}^{+\infty} \left(\frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \right) \hat{D}(\pm ae^{-i\phi}) \hat{S}(r) \nu(\mu) \hat{S}^\dagger(r) \hat{D}^\dagger(\pm ae^{-i\phi}) d\phi, \quad (98)$$

where $\pm a \in \mathbb{R}$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{DSTS}^{(\min)}(\rho_+, \rho_-) = \frac{1}{2} \left[1 - \frac{1}{2} \text{Tr} \left[\sqrt{(\rho_+ - \rho_-)^\dagger (\rho_+ - \rho_-)} \right] \right]. \quad (99)$$

The Homodyne detection probability distribution is

$$P_{DSTS}^{(HD)}(x|\pm) = \int_{-\infty}^{+\infty} P(\phi) |\langle x|\hat{D}(\pm ae^{-i\phi})\hat{S}(r)\nu(\mu)\hat{S}^\dagger(r)\hat{D}^\dagger(\pm ae^{-i\phi})|x\rangle| d\phi, \quad (100)$$

$$= \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \frac{1}{\sqrt{2\pi V(\phi, \beta, N, \mu)}} \exp \left[-\frac{\left(x \mp 2\sqrt{N(1-\beta) + \frac{\mu-1}{2\mu}(1+2N\beta)} \cos \phi \right)^2}{2V(\phi, \beta, N, \mu)} \right] d\phi, \quad (101)$$

where

$$V(\phi, \beta, N, \mu) = \frac{e^{-2\text{arcsinh}(\sqrt{N\beta})} \cos^2 \phi + e^{2\text{arcsinh}(\sqrt{N\beta})} \sin^2 \phi}{\mu}. \quad (102)$$

The error probability is

$$P_{\text{err}}^{\text{DSTS}}(N, \sigma, \mu, \beta) = \int_0^\infty P_{DSTS}^{(HD)}(x|-) dx \quad (103)$$

$$= \left[\frac{1}{2} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\phi^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}} \text{erfc} \left[\sqrt{\frac{2\left(N(1-\beta) + \frac{\mu-1}{2\mu}(1+2N\beta)\right)}{V(\phi, \beta, \mu, \nu)}} \cos \phi \right] d\phi \right]. \quad (104)$$